RESEARCH · BIOENERGY · R&D PARTNERSHIP PITCH · 2026

Extraction and Utilisation of Lignin as a Bio-Additive in a Compression Ignition Engine

Fuelled with diesel fuel — an approach to improve fuel properties.

A research pathway from managed biomass to measured fuel blends.

FEEDSTOCK Prosopis juliflora

STAGE Planning & validation

LOCATION KARE · Tamil Nadu

PRESENTED BY

Srinath Vatchavari Venkateshan

Team: Aarya Brat Shankar · R. Bhuvaneshwaran

Mentor: Dr. S. K. Gowthaman · Kalasalingam Academy of Research and Education

Research concept · Experimental results are not yet claimed

Research concept, IP & copyright: Srinath Vatchavari Venkateshan.

Website design & development: Srinath Vatchavari Venkateshan.

Why lignin, and why this feedstock?

Diesel remains central to transport, farming and industry. New fuel components must earn their place through measurement, not assertion.

01

THE CHALLENGE

Diesel is hard to displace, and alternative fuel components face strict quality, compatibility and engine requirements.

02

THE MATERIAL

Lignin is a carbon-rich aromatic polymer in plant cell walls. It can be separated and studied as a source of fuel-related products.

03

THE IDEA

Recover lignin from managed Prosopis juliflora, convert it into a liquid product, and test whether it can work as a diesel bio-additive.

MAIN QUESTION

Can a lignin-derived product be made stable and suitable for limited diesel blending — while meeting fuel-quality and engine requirements?

FEEDSTOCK BOUNDARY Use only Prosopis juliflora collected through authorised removal or management. The project must not encourage planting or spreading this invasive species.

What published research tells us

The literature supports investigation — not a performance claim.

2015 APPLIED ENERGY

Lignin as a stabiliser in fuel–water emulsions for a diesel engine. Useful background, but not the same as a lignin-derived pyrolysis liquid.

2024 SAE TECHNICAL PAPER

A lignin-based fuel tested in a small CI engine reported fuel-system and deposit challenges. Compatibility testing is essential.

2026 FUEL

Ignition-quality assessment of lignin-ethanol oil in a small CI engine. Supports further research; does not prove suitability for road diesel engines.

THE GAP

Our study must connect one selected feedstock, lignin recovery, product chemistry, diesel-blend stability and engine testing in one measured workflow.

Key lesson: lignin is not automatically a ready-to-use diesel additive. Separation, upgrading and compatibility checks come first.

A five-stage evidence chain

Each stage is designed to produce a record that the next stage can rely on.

01

RECOVER

Study a suitable method to recover lignin from authorised Prosopis juliflora biomass.

02

CHARACTERISE

Measure lignin yield, purity and key chemical features.

03

CONVERT

Study controlled conversion of recovered lignin into a liquid product.

04

SCREEN

Measure fuel properties and check blend stability before engine trials.

05

VALIDATE

Compare selected blends with diesel in a supervised CI-engine programme.

PROPOSED NOVELTY

A feedstock-specific, end-to-end evidence chain: recovered lignin → measured blend → supervised engine assessment.

BEFORE SUBMISSION

Novelty must be confirmed through a formal literature and patent search. The project remains exploratory.

From managed biomass to measured evidence

Eight proposed stages. None is complete at the time of publication; all measurements are planned references.

1 MANAGED BIOMASS

Prosopis juliflora, authorised and traceable

2 PREPARATION

Clean, dry, reduce size and control moisture

3 LIGNIN RECOVERY

Select and optimise the recovery method

4 CHARACTERISATION

Yield, purity and chemical structure

5 CONVERSION

Controlled conversion study

6 LIQUID PRODUCT

Separate and analyse the product

7 BLEND SCREENING

Stability and fuel-property gates

8 ENGINE EVALUATION

Only after safety and lab approval

Important: raw lignin and untreated conversion liquid are not assumed to be directly suitable for diesel blending.

What will be measured

The measurement plan is the core of the proposal: every later claim depends on these records.

STAGE	MEASUREMENTS	WHY IT MATTERS
Lignin	Recovery yield, purity, chemical structure	Checks whether the separation method works.
Liquid product	Density, viscosity, heating value, flash point, acidity, water, composition	Shows whether the product has useful fuel-related properties.
Blend screening	Visual separation, storage stability, viscosity, material compatibility	Checks whether the blend remains uniform and suitable for further tests.
Engine study	Performance, combustion, fuel use, emissions, deposits	Compares the selected blend with baseline diesel.

BLEND RANGE
Screening up to a maximum of 20% by volume. A research boundary — not a proven safe or optimal blend.

CONTROL
Baseline diesel as the reference. Only blends that pass laboratory screening proceed.

GATE
Engine testing requires a qualified lab, approved risk assessment and a supervised protocol.

No engine result, emissions reduction or fuel-economy improvement is claimed at this stage.

Every number is measured before it is claimed.

This is a research record, not a results announcement.

NOT CLAIMED TODAY

- × Emissions reduction
- × Engine-performance improvement
- × Commercial or policy benefit
- × Funding or partnership secured

WILL BE MEASURED

- Lignin yield and purity
- Product composition and fuel properties
- Blend stability and compatibility
- Performance, combustion and emissions (gated)

HOW WE PROTECT THE RECORD

- Documented methods and records
- Baseline comparison with diesel
- Qualified lab and approved risk assessment
- Results published only after review

Evidence gate: no result leaves the project until its method, baseline and review are documented.

Waste-to-value, to be evaluated

Impact is a hypothesis to test, not a headline to assert.

CIRCULAR USE

Could biomass residues support a more circular use of plant material? To be evaluated.

LOCAL FEEDSTOCK

Could a managed, locally available species provide a traceable feedstock? To be assessed.

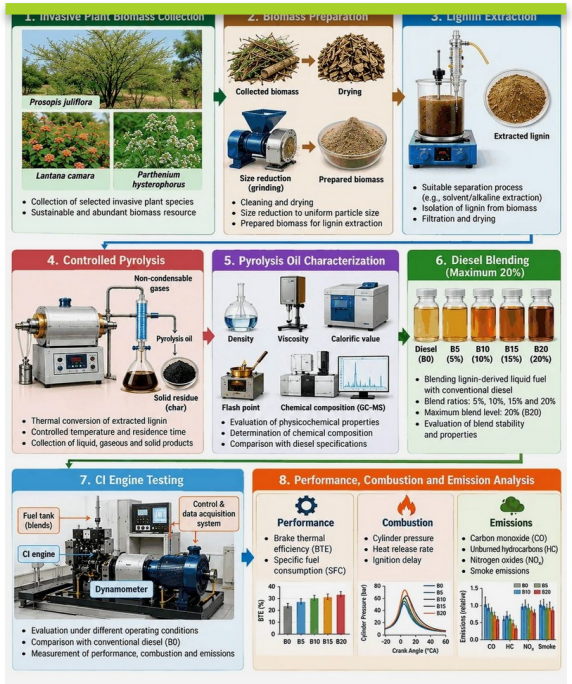
CAPABILITY

Builds methods, records and analytical capability for lignin-to-fuel research.

EVIDENCE

A documented feedstock-to-property dataset that later studies can scrutinise.

No commercial, market or policy benefit is claimed. Environmental claims require measurement and independent review.



Illustrative project imagery — not experimental results.

The partnership we need

One project, five kinds of support. Proposed partners only — no commitment is implied.

BIOMASS & ENVIRONMENT

Authorised feedstock access, ecological safeguards and traceability.

CHEMISTRY / MATERIALS LAB

Lignin recovery, purity, structure and analytical testing.

BIOENERGY / PROCESS LAB

Conversion route, product separation and upgrading research.

ENGINE TEST CENTRE

Fuel compatibility, supervised CI-engine evaluation and emissions.

INDUSTRY PARTNER

Technical mentoring, application needs, co-funding or in-kind support.

First step: a technical meeting and feasibility review. Formal collaboration will be documented separately.

A realistic route to research funding

Sequenced for a student-led project with faculty and laboratory support.

ANRF / DST — FIRE

Connects academic research with industry. The scheme page describes a 50% monetary match by DST-SERB. Requires an eligible academic partner and an industry partner; the academic lead must meet eligibility rules.

ANRF — ADVANCED RESEARCH GRANT

A competitive, investigator-led route for strong research questions in engineering, environment and related fields. A qualified faculty researcher should lead the application.

INSTITUTIONAL PARTNERSHIPS

Large ANRF catalytic programmes are institution-led and require substantial multi-partner commitments. They are not a direct student grant route.

RECOMMENDED SEQUENCE

01 Secure faculty PI + host lab

→ **02** Complete feasibility and baseline data

→ **03** Obtain industry / lab support letters

→ **04** Build a costed, milestone-based proposal

→ **05** Apply when an eligible call is open

Funding note: a million-USD-scale programme is a long-term consortium ambition only. It is not a current award, confirmed budget or guaranteed outcome.

From feasibility to validation

A staged plan with evidence gates — not a promise of results.

0–3 MONTHS

SCOPE & SETUP

Faculty PI, literature and patent review, feedstock permission, laboratory plan.

4–8 MONTHS

LIGNIN RECOVERY

Compare methods; record yield, purity and repeatability.

9–13 MONTHS

CONVERSION & ANALYSIS

Prepare liquid-product samples; analyse composition and key properties.

14–18 MONTHS

BLEND SCREENING

Stability, viscosity, compatibility and fuel-quality gates.

19–24 MONTHS

VALIDATION & SCALE-UP CASE

Supervised engine tests, LCA, techno-economics and next proposal.

BUDGET HEADS TO COST

- ✓ Feedstock collection & preparation
- ✓ Analytical testing & instrument access
- ✓ Blend stability & fuel-property tests
- ✓ Personnel, travel, contingency & overheads
- ✓ Extraction chemicals, glassware & consumables
- ✓ Conversion / upgrading facility access
- ✓ Engine test time, safety & emissions analysis

Budget to be costed after the PI, host laboratory and testing partners are confirmed.

Let's build the evidence together

A focused first conversation is the right next step — not a request for a commitment today.

WHAT WE ARE SEEKING

- ✓ An eligible faculty Principal Investigator
- ✓ A host laboratory for lignin recovery and analysis
- ✓ Access to fuel-property and blend-stability testing
- ✓ A qualified engine-testing collaborator
- ✓ An industry mentor and future funding partner

FIRST STEP

A technical meeting and feasibility review.

Email: **9926009061@klu.ac.in**

KARE · Krishnankoil, Tamil Nadu · github.com/vvsrinath

PROJECT OWNERSHIP & CREDIT

Research concept, direction, outputs, documents, intellectual property and copyright are owned and managed solely by **Srinath Vatchavari Venkateshan**.

Aarya Brat Shankar and R. Bhuvaneshwaran are acknowledged supporting contributors. Dr. S. K. Gowthaman advises as faculty mentor. Acknowledgement, authorship credit or mentorship does not confer ownership, copyright or other rights.

Website design & development: **Srinath Vatchavari Venkateshan**.

Any separate written agreement signed by the owner takes precedence.

Evidence trail

Selected references and official funding sources. Full abstracts and poster are available on the project site.

01	**Terauchi et al. (2024).**	Experimental Study of Lignin Fuels for CI Engines. SAE Technical Paper. doi:10.4271/2024-37-0022
02	**Fuel (2026).**	Ignition quality assessment of alternative marine fuels using a small CI engine, 413, 138117.
03	**Applied Energy (2015).**	Performance, combustion, and emissions in a diesel engine operated with fuel-in-water emulsions based on lignin. doi:10.1016/j.apenergy.2015.05.036
04	**Yang et al. (2007).**	Characteristics of hemicellulose, cellulose and lignin pyrolysis. Fuel, 86, 1781–1788. doi:10.1016/j.fuel.2006.12.013
05	**ANRF / DST-SERB.**	Fund for Industrial Research Engagement (FIRE) and Advanced Research Grant information. Verify live call dates and eligibility before applying.

Sources checked 1 November 2026. Verify live call dates, eligibility and applicable rules before submission.